

TRANSPORT PHENOMENA

ChE 319 – Spring 2023

Unique # - 15555

Class Meeting: CPE 2.222 on MWF from 1:00 pm – 2:00 pm

Instructor: Prof. Gabriel E. Sanoja (he/him), EER 5.618, gesanoja@che.utexas.edu

The subject of transport phenomena includes three closely related topics: fluid dynamics, heat transfer, and mass transfer. Fluid dynamics involves transport of *momentum*, heat transfer deals with the transport of *energy*, and mass transfer is concerned with the transport of *mass* of various chemical species.

TEXTBOOKS:

Required: R. B. Bird, W. E. Stewart and E. N. Lightfoot, *Transport Phenomena*, 2nd Ed., John Wiley & Sons, Inc., 2002.

Optional: W. M. Deen, *Analysis of Transport Phenomena*, 2nd Ed., Oxford University Press., 2012.

Optional (Fluid Mechanics): M. M. Denn, *Process Fluid Mechanics*, Prentice-Hall International Series in the Physical and Chemical Engineering Sciences, 1980

Optional (Mathematics): J. Stewart, *Calculus*, 8th Ed., Cengage Learning, 2015.

PREREQUISITES:

ChE 317-Introduction to Chemical Engineering Analysis and M 427K-Advanced Calculus for Applications I.

TEACHING ASSISTANT:

Tiffany Jeng, tjeng@utexas.edu

TUTOR:

Keyu Chen, keyuchenacademic@gmail.com

RECITATION:

W 4:00 - 6:00 p.m. CPE 2.222

OFFICE HOURS:

TTh 5:00 – 6:00 pm Sanoja – EER 5.702
MT 4:00 – 5:00 pm Jeng – EER 5.702

TUTORING:

MT 3:00 – 4:00 pm Chen – EER 5.702

PARTICIPATION:

There will be an assessment of your engagement and participation in the course which will represent 5% of your letter grade. **You are expected to attend lecture, recitation, and some office hours; arrive on time; turn off and stow away electronic devices; and ask questions. Another way to actively participate in the course is to submit a detailed solution to one (1) homework problem, either in LaTeX or Microsoft Word, through Canvas by the assigned deadline.**

QUIZZES:

There will be eleven (11) quizzes administered throughout the semester at the beginning of recitation. I will disregard your lowest quiz grade. Please be aware that missed quizzes will receive no credit and cannot be made up. Quizzes account for 20% of your grade.

HOMEWORKS:

There will be homework nearly every week. These will be composed of traditional mathematically-oriented problems, but also of some readings and videos related to soft-skills and grand societal challenges in engineering. Homework will not be collected nor graded, but will be the basis of the quizzes. **You are expected to actively learn by doing homework so that you can score perfectly on the quizzes. Also, please note that you can submit a detailed solution to one (1) homework problem, either in LaTeX or Microsoft Word, through Canvas by the assigned deadline to participate in the course.**

MIDTERMS AND FINAL:

There will be two (2) midterms administered during recitation, and one (1) final. These will be closed book and notes, but you will be allowed one (1) US Letter size-paper with content of your choice as supportive material. Please be aware that missed exams cannot be made up. If you miss one (1) midterm due to very exceptional circumstances, you will receive the weighted-grade of the final. If you miss both midterms, you will be referred to the Deans Office (ECJ 2.200) and subject to administrative proceedings. If you do not miss any midterm, your lowest midterm-grade will be substituted with that of the weighted-grade of the final in as long as this is beneficial for your letter grade.

If you want to request a regrade of your exam, please write an explanatory note, attach it to your exam, and submit it directly to me within one class period. Failure to comply with this procedure could result in denial of your request. Please be aware that the entire exam will be regraded.

EVALUATION:

Quizzes*	20% - 2% each
Midterms [#]	40% - 20% each
Participation	5%
Final	35%

*Lowest grades will be disregarded. Missed quizzes will receive no credit and count towards the lowest grades.

[#]You may substitute the lowest grade of a Midterm with that of the Final, properly weighted, as long as (i) you do not miss any Midterm and (ii) this substitution is beneficial for your final letter grade.

Letter grades will be determined using the following scale:

A	A ⁻	B ⁺	B	B ⁻	C ⁺	C	C ⁻	D ⁺	D	D ⁻	F
100-90	89-85	84-80	79-75	74-70	69-66	65-63	62-60	59-56	55-53	52-50	49-0

Please be cautious when using the average calculated by Canvas to determine your standing in the course. This is calculated automatically and is not always accurate.

If you fail to score nearly perfectly on the first two quizzes, you are strongly encouraged to drop the class. **You are expected to actively study from the very beginning of the semester to perform well in this course. Studying *actively* entails attending lecture, recitation, and office hours; as well as critically doing the homework.** The instructor, TA, and tutor are available in Office Hours and Tutoring to clarify concepts and help you master problem-solving skills.

ABET OUTCOME:

An ABET (Accreditation Board for Engineering and Technology) outcome of the course will be for students to have the **“ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.”** AIChE (American Institute for Chemical Engineers) defines a complex engineering problem as having one or more of the following characteristics: **“involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.”**

ACCOMMODATIONS:

The University of Texas at Austin is committed to create an accessible and inclusive learning environment consistent with university policy, and federal and state law. **If you experience any learning barriers, please let me know so that I can work to ensure you have equal opportunity to participate and succeed in this course.** If you are a student with a disability and need accommodation, please contact [Services for Students with Disabilities \(SSD\)](#). If you are already registered with SSD, please submit your Accommodation Letter directly to me as early in the semester as possible.

PRONOUNS:

Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with differences of race, culture, religion, politics, sexual orientation, gender, gender variance, and nationalities. Class rosters are provided to the instructor with the student's legal name, unless they have added a "preferred name" with the [Gender and Sexuality Center](#). I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use (she/he/they/ze, etc). Please advise me of any changes early in the semester so that I can update my records. You are also welcome to [add your pronouns to Canvas](#).

ACADEMIC INTEGRITY:

Each student in the course is expected to abide by The University of Texas Honor Code: "As a student at The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity". **You are responsible for understanding [UT's Academic Honesty and the University Honor Code](#).**

TITLE IX REPORTING:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university's relevant policies.

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, to report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee.** If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, contact the [Title IX Office](#) via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as sexual misconduct, including the types of sexual misconduct covered under Texas Senate Bill 212. The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.

IMPORTANT DATES (See also [Academic Calendar 2022-2023](#))

Jan. 09 – M. First class in CPE 2.222 from 1:00 pm to 2:00 pm
Jan. 11 – W. First recitation in CPE 2.222 from 4:00 pm to 6:00 pm.

Jan. 12 – Th.	Last day to add a class without permission.
Jan. 16 – M.	Martin Luther King, Jr. Day; no class held.
Jan. 18 – W.	Quiz #1 in CPE 2.222 from 4:00 pm to 4:30 pm.
Jan. 25 – W.	Quiz #2 in CPE 2.222 from 4:00 pm to 4:30 pm. Also, last day to drop a class without permission.
Feb. 01 – W.	Quiz #3 in CPE 2.222 from 4:00 pm to 4:30 pm.
Feb. 08 – W.	Quiz #4 in CPE 2.222 from 4:00 pm to 4:30 pm.
Feb. 15 – W.	Midterm #1 in CPE 2.222 from 4:00 pm to 6:00 pm.
Feb. 22 – W.	Quiz #5 in CPE 2.222 from 4:00 pm to 4:30 pm.
Mar. 01 – W.	Quiz #6 in CPE 2.222 from 4:00 pm to 4:30 pm.
Mar. 08 – W.	Quiz #7 in CPE 2.222 from 4:00 pm to 4:30 pm.
Mar. 13 – 18.	Spring Break
Mar. 21 – T.	Last day to Q-drop a class or change a class to pass/fail.
Mar. 22 – W.	Quiz #8 in CPE 2.222 from 4:00 pm to 4:30 pm.
Mar. 29 – W.	Midterm #2 in CPE from 4:00 pm to 6:00 pm.
Apr. 05 – W.	Quiz #9 in CPE 2.222 from 4:00 pm to 4:30 pm.
Apr. 12 – W.	Quiz #10 in CPE 2.222 from 4:00 pm to 4:30 pm.
Apr. 19 – W.	Quiz #11 in CPE 2.222 from 4:00 pm to 4:30 pm.
Apr. 24 – M.	Last class day.
TBD.	Final Exam.

This syllabus is from January 9th, 2023 and is subject to change. Students who miss class are responsible for learning about any changes to the syllabus.

COURSE OUTLINE:

The course is composed of four (4) modules and forty-four (44) lectures; some of which should be complemented with book readings.

Module 1: Energy Transport

1. Syllabus. The subject of transport phenomena. Mechanisms of energy transport. **BSL 0, 9.1, 9.7.**
2. Conservation of energy. Fourier's law of heat conduction. **BSL 10.1, 11.1-11.2.**
3. 1-D steady-state conduction. Thermal circuits. **BSL 10.6.**
4. Heat conduction with thermal energy generation. **BSL 10.2-10.5**
5. Dimensional analysis and Biot number.
6. 2-D steady-state conduction. **BSL 10.7.**
7. Fins.
8. Transient conduction. **BSL 12.1**
9. Lumped capacitance method.
10. Pseudo-steady state conduction. Phase transitions.

Module 2: Mass Transport

11. Mechanisms of mass transport. Mixture velocity. **BSL 17.1, 17.7-17.8.**
12. Conservation of chemical species. Fick's law of diffusion. **BSL 18.1, 19.1-19.2.**
13. 1-D steady-state diffusion. Mass transfer circuits.
14. Diffusion with chemical reaction. **BSL 18.3-18.5**
15. Dimensional analysis and Damköhler number.
16. Diffusion through a stagnant gas. **BSL 18.2**
17. Pseudo-steady state diffusion. Membranes.
18. Diffusion in semi-infinite membranes. Boundary layer. **BSL 20.1**
19. Similarity solution. Error function.

Module 3: Momentum Transport (Fluid Mechanics)

20. Mechanisms of momentum transport. Density. **BSL 1.1-1.2, 1.7.**
21. Viscosity. Newton's law of viscosity.
22. Conservation of mass. Substantial derivative. **BSL 3.1**
23. Conservation of momentum. **BSL 2.1, 3.2, 3.3, 3.5.**
24. Hydrostatic and modified pressure.
25. Couette flow. Navier-Stokes equation.
26. Hagen-Poiseuille flow. **BSL 2.2-2.5, 3.6.**

- 27. Dimensional analysis. Reynolds number and friction factor. **BSL 3.7, 11.5, 19.5.**
- 28. Stokes' flow. Streamlines. **BSL 2.6, 4.2.**
- 29. Stokes' flow. Velocity, shear stress, and pressure.
- 30. Stokes' flow. Form and friction drag. Drag coefficient.

Module 4: A Unified Picture

- 31. Forced convection. Péclet number. **BSL 11.4, 19.4.**
- 32. Forced convection at large Pe.
- 33. Bulk concentration. Mass exchangers. **BSL 22.1-22.2.**
- 34. Overall mass transfer coefficient. Sherwood number.
- 35. Mass transfer in entrance region.
- 36. Sherwood number in entrance region.
- 37. Mass transfer in fully-developed region.
- 38. Sherwood number in fully-developed region.
- 39. TBD
- 40. TBD
- 41. TBD
- 42. Summary. Course evaluations